Ex.no: 10 Date: 10/9/26

Internetworking with Routers in Cisco Packet Simulator.

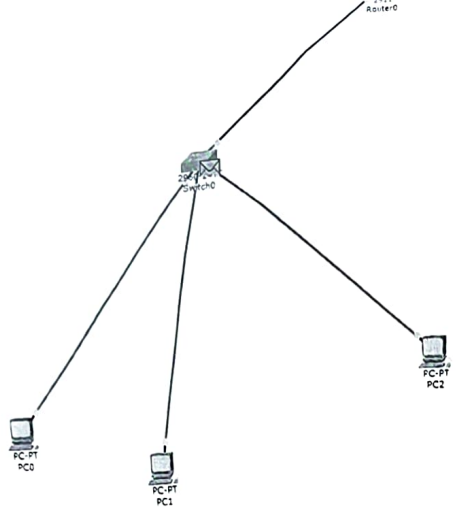
a) Simple Internetwork using a Router.

* Aim:

To Design and configure a simple internetwork using a router in Cisco Packet Tracer and observe the routing of Packets between network.

* Procedure:

1) Open Cisco packet Tracer.
2) Place one Cisco 2911 Router, two Switch and 3-PC for each network.
3) Connect the PCs the switch using copper straight cables.
4) Connect the switches to the Router using Ethernet Cables.
5) Assign static IP addresses to all PCs and router interface.
6) Configure the router interface using assigned IP addresses.
7) Enable the router interface using no shutdown command.
8) Verify the interface status using show ip interface brief
9) Using simple PDU to sent packet from one network to other.
10) Select simulator mode and observe the packet transfer.




```
--- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

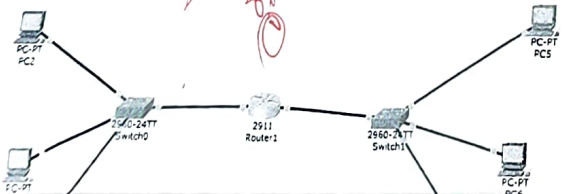
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

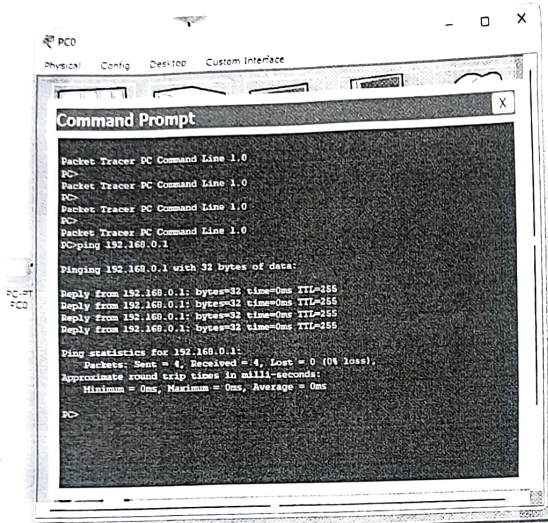
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up
exit
```

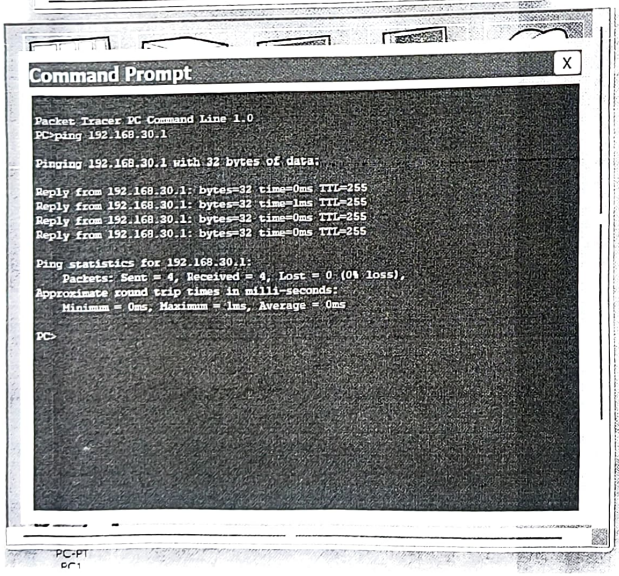
```
Router(config)#interface gigabitethernet 0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up
exit
```



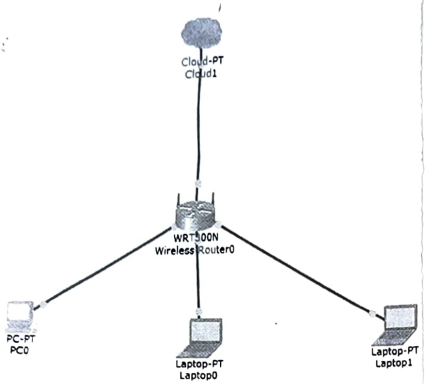


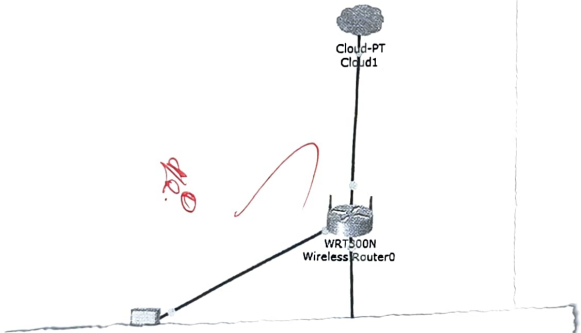


11) observe what the router forwards the packet through the appropriate interface according to destination network.

* Result!

The simple internetwork was successfully design and configured using a router in CPT. Packet Transmission and router between different network were successfully.





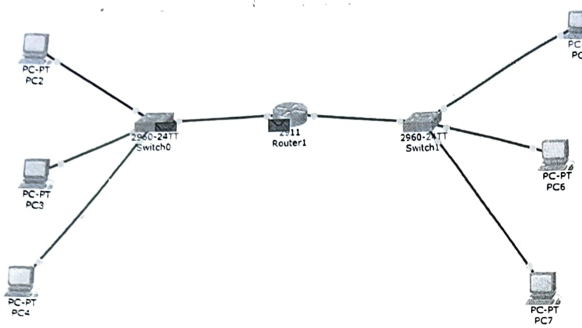
6) Wireless Router with DHCP Server and Internet Cloud.

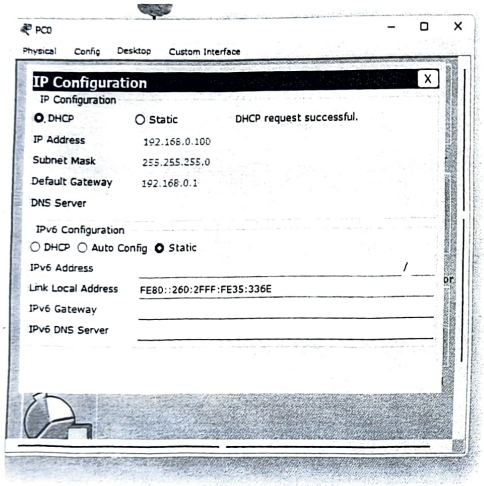
* Aim:

1.) Open Cisco Packet Tracer.
2.) Place one cloud-PT, one WRT 300N Wireless Router, one PC and two laptop.
3) Connect the cloud-PT to the WAN Port of WRT 300N using copper Straight Through Cable.
4) Connect the pc to a LAN Port of the WRT 300N using a copper Straight LAN Cable.
5) Install a wireless Card WPC 300N in the Laptops if required.
6) Configure the WRT300N with AP IP address 192.169.01. and subnet mask 255.255.255.0
7) Enable the DHCP Server in wireless router.
8.) Connect the Laptops to the received IP address automatically ~~successfully~~.
9.) Verify that the devices have received IP address successfully.
10.) Use the ping command to test connect between the device.
11.) Use simulator mode to observe the packet transmission through the wireless router.

* Result:

The internetwork using a wireless router, DHCP Server and internet cloud was successfully designed and configured. The devices obtained IP config automatically through DHCP and communication btw the devices was successfully verified.





c) Internetwork Using Switch, Router and Ethernet cloud.

* Aim:

To Design and configure an interface using Switch router and cables in Cisco packet Tracer.

* Procedure:

1) Open Cisco Packet Tracer.
2) Place one Cisco 2911 Router one Switch and 3PCs
3) Connect the router to the switch using a copper straight through cables.
4) Connect the three PCs to different port of the switch using straight through cables.
5) Assign an IP address to the router interface.
6) Assign an IP address to all PCs
7) Configure the default gateway of each PC using Router.
8) Enable the router using the no shutdown IP address.
9) Check the router interface status using show IP interface brief.
10) Test Connectivity between PCs using Ping Command.
11) Using Simulate mode and observe the packet travelling through switch and router.

* Result:

The internetwork using a router, switch and Ethernet cables was successfully designed and config in CPT Communicated between the connect devices and successfully verified using the Ping Command.